

Environmental Factors

At the end of this episode, I will be able to:

1. Environmental factors

Leamer Objective: *Learn about the various environmental factors you should consider with physical IT installations*

Description: In this episode, you will learn about key environmental considerations you should have when it comes to physical IT installations. These considerations include humidity, fire suppression, and temperature.

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- Humidity is a crucial environmental factor in modern IT installations, including data centers, server rooms, and other facilities housing sensitive electronic equipment. Proper humidity levels are essential for maintaining the reliability and performance of IT infrastructure. Humidity refers to the amount of moisture or water vapor present in the air. It is often expressed as a percentage, representing the relative humidity (RH) in relation to the maximum amount of moisture the air can hold at a given temperature. The recommended humidity range for IT installations is typically between 40% and 60% RH. This range helps strike a balance between preventing electrostatic discharge (ESD) and avoiding excessive moisture that could lead to corrosion.
 - Fire suppression is a critical aspect of environmental considerations in IT installations, such as data centers, server rooms, and other facilities housing electronic equipment. The goal of fire suppression systems is to detect and extinguish or control fires quickly, minimizing damage to equipment and ensuring the safety of personnel. The main types of fire suppression systems in use today include water-based systems, gaseous suppression systems, and dry chemical systems. Gaseous suppression systems are preferred in IT installations because they are clean agents that do not leave residue. This minimizes the potential for damage to electronic equipment and simplifies the cleanup process. When selecting fire suppression systems, environmental considerations include the impact of the suppression agents on the atmosphere, ozone layer, and overall environmental sustainability.
 - Temperature is a critical environmental factor in IT physical installations, encompassing data centers, server rooms, and other facilities hosting electronic equipment. Maintaining optimal temperature levels is essential for ensuring the reliable operation, performance, and longevity of IT infrastructure. IT equipment manufacturers typically provide specifications indicating the recommended temperature range for their products. This range is crucial for optimal performance and reliability. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) publishes guidelines for data center environmental conditions, including temperature recommendations.
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Additional Resources:

Physical and Environmental Security

<https://csrc.nist.gov/publications/nistpubs/800-12/800-12-html/chapter15.html>